Quantum Teleportation: From Alice to Bob

Simulation and Execution on IBM Quantum Hardware

Field	Details
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Event	IBM Qiskit Fall Fest 2026 at RV University
Date	17 October 2026

1. Objective

To implement the quantum teleportation protocol, verify it on the Qiskit Aer simulator, and run it on real IBM Quantum processors, comparing ideal behaviour against hardware noise.

2. Background

The No-Cloning Theorem states that an unknown quantum state cannot be copied, and measuring it directly destroys it. Quantum teleportation (Bennett et al., 1993) gets around this by transferring the state of a qubit using one shared entangled pair and two classical bits.

The protocol works as follows:

- Alice and Bob share a Bell pair $(|00\rangle + |11\rangle)/\sqrt{2}$.
- Alice applies a CNOT (message qubit → her half of the pair), then a Hadamard on the message qubit, and measures both qubits.
- She sends the two classical bits (c0, c1) to Bob.
- Bob applies the correction Z^c0 X^c1 to his qubit, which recovers Alice's original state exactly.

3. Implementation

The project is a full-stack application with a Qiskit and Qiskit Aer quantum engine, a FastAPI backend and a React frontend. Features include:

- Interactive 3D Bloch sphere and a 7-stage protocol walkthrough.
- Multi-shot Aer simulation (512 to 8,192 shots) with a chi-square test against the expected uniform distribution of the four outcomes.
- Hardware noise sweep using calibration data from ibm_kingston (156-qubit Heron r2).
- Live IBM Cloud queue status and job submission through qiskit-ibm-runtime (SamplerV2), with circuits transpiled to the native gate set (cz, rz, sx, x).
- Automated verification across 8 cardinal states, backed by 36 unit tests.

4. Execution on IBM Quantum Platform

The teleportation circuit was transpiled and submitted to physical IBM Quantum backends. The jobs are shown below.

Backend	Job ID	Date / time	Shots	Status
ibm_kingston	db9i2agc77vs7381hb50	17 Oct 2026, 12:48 PM	1024	Completed
ibm_kingston	db9hc4hgv2cc7…	17 Oct 2026, 12:00 PM		Completed
ibm_marrakesh	db70ur04qg6s7…	13 Oct 2026, 4:31 PM		Completed

Figure 1 lists the workloads on the IBM Quantum Platform. Figure 2 shows the details of the latest job: it ran on ibm_kingston using the sampler primitive, completed in 17 s with 2 s of quantum usage. Figure 3 shows the measurement histogram over 1024 shots, and Figure 4 shows the transpiled circuit that was executed.

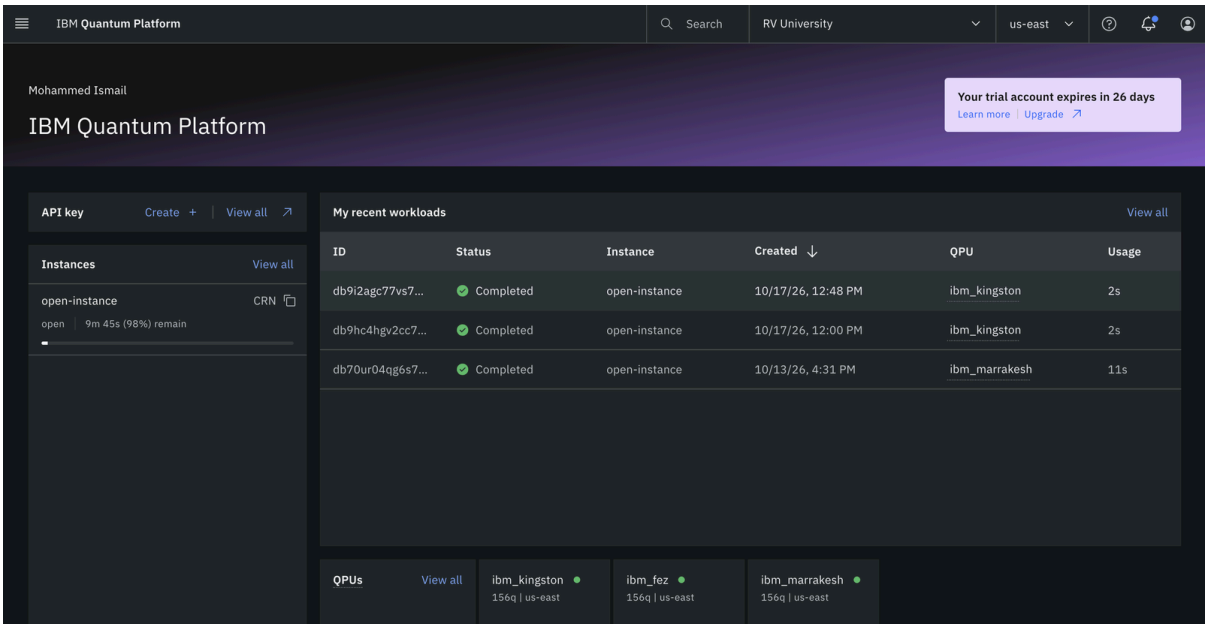


Figure 1: IBM Quantum Platform recent workloads (ibm_kingston and ibm_marrakesh)

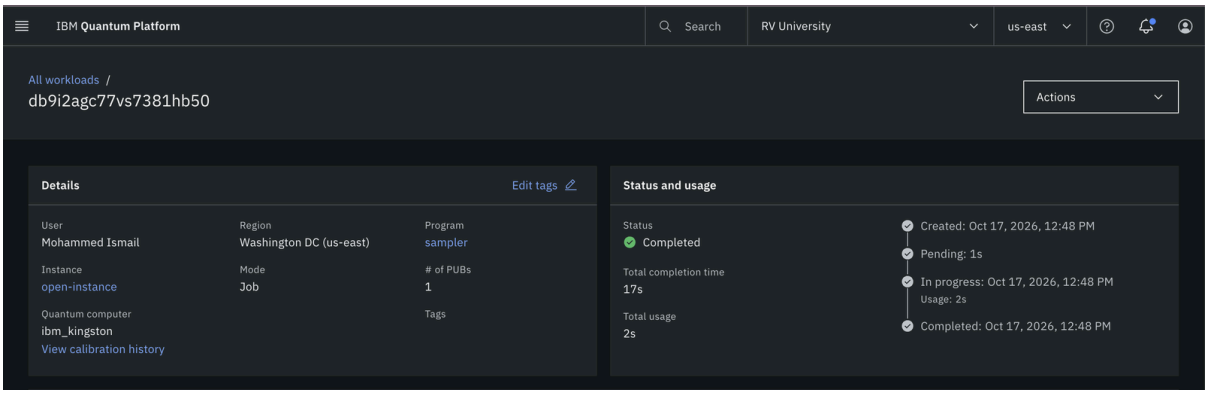


Figure 2: Job db9i2agc77vs7381hb50 details on ibm_kingston

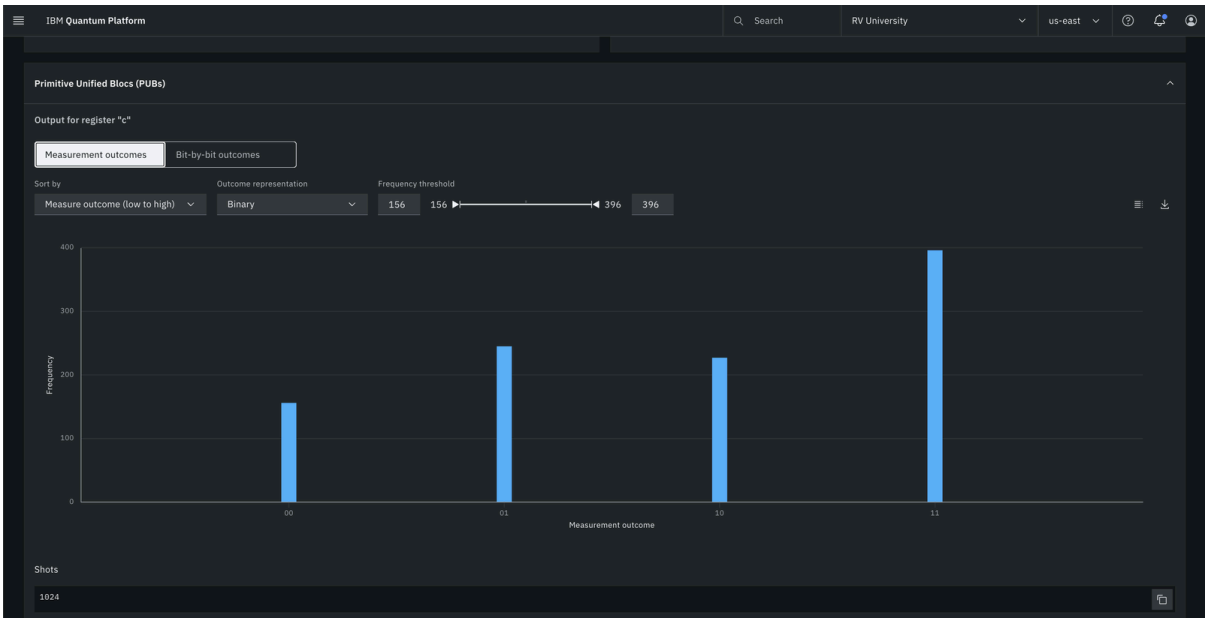


Figure 3: Measurement outcomes for register c (1024 shots)

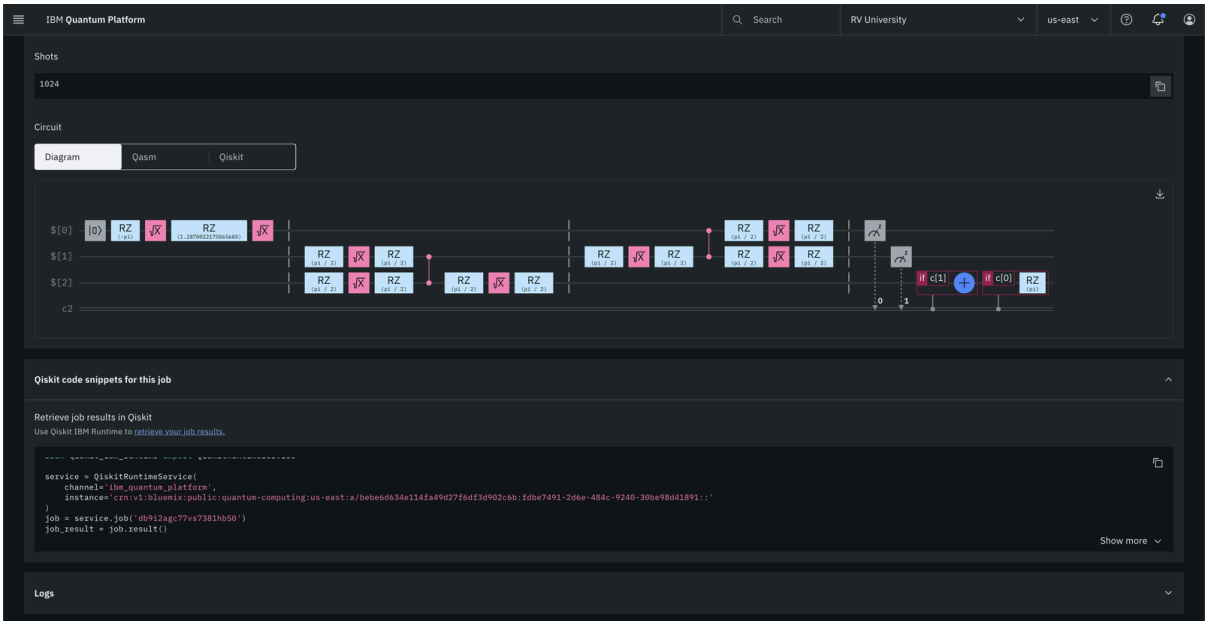


Figure 4: Transpiled teleportation circuit with mid-circuit measurement and feed-forward

5. Results and Observations

- Simulator: all four measurement outcomes (00, 01, 10, 11) occur with probability close to 25%, and Bob's corrected state has fidelity 1.000000 in every branch.
- Hardware: calibration for ibm_kingston gave T1 ≈ 232.01 µs, T2 ≈ 136.34 µs, CZ gate error ≈ 0.196% and readout error ≈ 0.995%, for an expected fidelity of about 99.41%.
- Hardware run (ibm_kingston, 1024 shots): the four outcomes ranged from 156 counts (00) to 396 counts (11) instead of the ideal 256 each. This clear deviation from a uniform distribution is attributed to gate, readout and decoherence noise on the device, plus any bias from the transpiled circuit.
- The circuit executed on hardware is a dynamic circuit: Alice's two qubits are measured mid-circuit, and Bob's X and Z corrections are applied conditionally on those results (the "if c[1]" and "if c[0]" blocks in Figure 4).

6. Conclusion

Quantum teleportation was verified in simulation, where it reproduces the input state with unit fidelity, and executed on IBM Quantum hardware. The hardware run completed successfully with the full feed-forward correction, and its outcome counts deviated from the ideal because of device noise.

7. Links

Source code: github.com/mohammedismailmi/quantum-teleportation

Live app: quantum-teleportation-qae8.onrender.com

References

1. Bennett, C. H., et al. (1993). Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels. Physical Review Letters, 70(13), 1895.

2. Wootters, W. K., & Zurek, W. H. (1982). A single quantum cannot be cloned. Nature, 299, 802–803.

3. Nielsen, M. A., & Chuang, I. L. (2010). Quantum Computation and Quantum Information. Cambridge University Press.

4. IBM Quantum and Qiskit documentation, docs.quantum.ibm.com.